

Experiment 7

Name: Urvi Shivalkar

PRN: B25ET1196

Class: SY-3(C)

Title: Evaluation of postfix expression.

Code:

```
#include <ctype.h>
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define MAX 100
```

```
int stack[MAX];
```

```
int top = -1;
```

```
void push(int val) {
```

```
    if (top >= MAX - 1) {
```

```
        printf("Stack overflow\n");
```

```
        return;
```

```
    }
```

```
    stack[++top] = val;
```

```
}
```

```
int pop() {
```

```
    if (top < 0) {
```

```
        printf("Stack underflow\n");
```

```
        exit(1);
```

```
    }
```

```
    return stack[top--];
```

```
}
```

```
int main() {
```

```
    char ch;
```

```
    printf("Enter a postfix expression (e.g., 5 2 3 * +): ");
```

```
    while (scanf("%c", &ch) == 1 && ch != '\n')
```

```

{

    if (ch == ' ' || ch == '\r') {
        continue;
    }

    if (isdigit(ch)) {
        push(ch - '0');
    }

    else {
        int val2 = pop();
        int val1 = pop();

        switch (ch) {
            case '+':
                push(val1 + val2);
                break;
            case '-':
                push(val1 - val2);
                break;
            case '*':
                push(val1 * val2);
                break;
            case '/':
                if (val2 == 0) {
                    printf("Error: Division by zero\n");
                    exit(1);
                }
                push(val1 / val2);
                break;
            default:
                printf("Invalid character/operator: %c\n", ch);
                exit(1);
        }
    }
}

```

```
// Print the final result
printf("Result is: %d\n", pop());

return 0;
}
```

Code:

Output

```
Enter a postfix expression (e.g., 5 2 3 * +): 987*4+7/-
Result is: 1
```